Lab 3, Act 1

Trusting a positive detection

NRES 746

Answer key. Answers are in magenta.

The scenario

A regional wildlife agency is monitoring a rare forest carnivore using motion-triggered camera traps. Most trigger events aren't the target species at all (deer, rabbits, blowing branches, and hikers set off the cameras far more often than the target animal). To keep field biologists from having to review every single image by eye, the agency uses an automated image classifier that flags each trigger event as either "target species detected" or not.

The classifier isn't perfect. For this exercise, suppose:

Quantity	Value
Prevalence: fraction of all trigger events that are actually the target species	1 in 10,000 (0.0001)
Sensitivity: P(flagged | actually the species)	0.98
False positive rate: P(flagged | not the species)	0.01

That sounds like a pretty good classifier, since it catches 98% of true detections and only mistakenly flags a "background" image not containing the species 1% of the time. **If the classifier flags an image, how worried should the field team be that it's a false alarm?**

You haven't been given a formula for this yet, but let's see if we can figure it out together!

Part 1: Count your way to an answer (individual, ~10 min)

Formulas are easier to trust once you've built one yourself. Start instead with plain counting. Suppose the cameras log $N = 100{,}000$ trigger events over a season.

1a. Fill in the table below using only the rates given above. Every blank cell is just one of those rates applied to a total you already know (e.g., how many of the 10 true-species events do you expect the classifier to flag, given its sensitivity?). Round to the nearest whole number.

	Flagged	**Not flagged**	**Total**
Actually the species	10	0	10
Not the species	1,000	98,990	99,990
Total	1,010	98,990	100,000

1b. Of the images the classifier *flagged*, what fraction are actually the target species? No formula needed – just use the counts in your completed table.

10/1,010 ≈ 0.0099, about 1%.

1c. Is this number close to what you expected, given how good the classifier's sensitivity and false-positive rate sounded at the start? In a sentence or two, what would you tell the field team about how to handle a single flagged image?

No, far lower than the 98% and 1% rates suggest. A single flagged image is almost always a false alarm, so confirm it (a second detection, sign, or a person checking the image) before acting.

Part 2: Turn your counting into a general recipe (individual, ~10 min)

Your table only works for exactly 100,000 events. Let's turn the same reasoning into something that works for *any* number of events.

2a. Rewrite the two counts you used in 1b – the number of flagged images that are truly the species, and the total number of flagged images – as expressions using N and the rates given at the top (prevalence, sensitivity, false positive rate), instead of plugged-in numbers.

True flags: $N \times \text{prevalence} \times \text{sensitivity}$. All flags: $N \times \text{prevalence} \times \text{sensitivity} + N \times (1 - \text{prevalence}) \times$ false positive rate.

2b. Divide your two expressions from 2a to reconstruct the fraction from 1b. N should appear in both the top and bottom – cancel it out. What are you left with?

$$\text{PPV} = \frac{\text{prev} \times \text{sens}}{\text{prev} \times \text{sens} + (1 - \text{prev}) \times \text{FPR}} = \frac{0.0001 \times 0.98}{0.0001 \times 0.98 + 0.9999 \times 0.01} \approx 0.0097$$

2c. State your result from 2b as a general rule, in your own words or symbols: given a classifier's sensitivity, false-positive rate, and the prevalence of the thing it's looking for, how do you compute the chance that a flagged event is real?

Chance a flag is real = true flags / all flags, with true flags = prevalence × sensitivity and false flags = (1 − prevalence)× false positive rate. (This is Bayes' rule.)

Part 3: Try it somewhere else (individual, ~5 min)

A second survey site monitors the same species with the *same* classifier (same sensitivity and false-positive rate), but the species is well-established there rather than rare: prevalence = 0.01 (1 in 100 trigger events), instead of the original site's 1 in 10,000.

3a. Using your recipe from Part 2 (or a fresh counting table, if you'd rather), find this second site's PPV – the chance that a flagged image is real.

$$\frac{0.01 \times 0.98}{0.01 \times 0.98 + 0.99 \times 0.01} \approx 0.497, \text{ about } 50\%.$$

3b. Same classifier, same sensitivity, same false-positive rate, but a different answer. What does comparing these two sites tell you about how much those two classifier properties matter versus how common (prevalent) the thing you're looking for actually is? We haven't talked about it yet, but feel free to use the term "base rate" to describe the prevalence (e.g., of the target predator) and speculate on how your findings today relate to a common logical fallacy known as the "base rate fallacy".

The classifier didn't change, but the chance a flag is real went from about 1% to about 50%: the base rate matters as much as the classifier's accuracy. The base rate fallacy is judging a positive result by the test's accuracy alone, ignoring how rare the target is.

Group discussion and revision (~5 min)

Compare your Part 2 recipe and your Part 3 answer with your group. Does everyone's recipe agree once you strip away notation differences? If not, find where they diverge.

Feel free to revise any of your answers above based on the discussion.

Please take a minute to record any changes you made and any new insights gained during the group discussion. You can also use this space to reflect on any areas of confusion ("muddiest points") or topics you want to review or dust off.

Open-ended.